

ASSESSMENT CO3 AT2

Fixed-Point vs Floating-Point Beamforming

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%% Fixed-Point vs Floating-Point Beamforming

clc; clear; close all;

% Parameters
c=3e8; f=28e9; lambda=c/f; d=lambda/2;
N=8; theta0=30; theta=-90:0.2:90;
n=0:N-1;

%% Floating-Point Beamforming
w=exp(-1j*2*pi*d*n*sind(theta0)/lambda);
AFf=zeros(size(theta));
for k=1:length(theta)
    a=exp(-1j*2*pi*d*n*sind(theta(k))/lambda);
    AFf(k)=abs(w'*a);
end
AFf=AFf/max(AFf);

%% Fixed-Point Beamforming
scale=2^10;
wf=round(real(w)*scale)/scale + 1j*round(imag(w)*scale)/scale;
AFx=zeros(size(theta));
for k=1:length(theta)
    a=exp(-1j*2*pi*d*n*sind(theta(k))/lambda);
    af=round(real(a)*scale)/scale + 1j*round(imag(a)*scale)/scale;
    AFx(k)=abs(wf'*af);
end
AFx=AFx/max(AFx);

%% Beam Pattern
figure;
plot(theta,20*log10(AFf),'b','LineWidth',2); hold on;
plot(theta,20*log10(AFx),'r--','LineWidth',2);
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grid on;

xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');
title('Floating-Point vs Fixed-Point Beamforming');
legend('Floating Point','Fixed Point');

ylim([-50 5]);

%%% Quantization Error

err=abs(AFf-AFx);

figure;

plot(theta,err,'k','LineWidth',2);

grid on;

xlabel('Angle (Degrees)');
ylabel('Magnitude Error');
title('Quantization Error');

%%% Error Statistics

MAE=mean(err);

MAXE=max(err);

RMSE=sqrt(mean(err.^2));

fprintf('\n=== Quantization Error ===\n');

fprintf('Mean Absolute Error = %.8f\n',MAE);

fprintf('Maximum Error = %.8f\n',MAXE);

fprintf('RMSE = %.8f\n',RMSE);

fprintf('\n=== Computational Comparison ===\n');

fprintf('Floating Point: High precision, higher memory, higher computational cost.\n');

fprintf('Fixed Point: Lower precision, lower memory, faster hardware implementation.\n');

fprintf('\n=== Recommendation ===\n');

if RMSE<0.01

    fprintf('Fixed-point implementation provides nearly identical performance.\nPreferred for FPGA/ASIC and
real-time communication systems.\n');

else

    fprintf('Floating-point implementation is preferred due to lower quantization error.\n');

end

```

Command Window Output

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Command Window

=== Quantization Error ===
Mean Absolute Error = 0.00007631
Maximum Error = 0.00032438
RMSE = 0.00009537

=== Computational Comparison ===
Floating Point: High precision, higher memory, higher computational cost.
Fixed Point: Lower precision, lower memory, faster hardware implementation.

=== Recommendation ===
Fixed-point implementation provides nearly identical performance.
Preferred for FPGA/ASIC and real-time communication systems.
>>
```

Figure 1: Beam Pattern

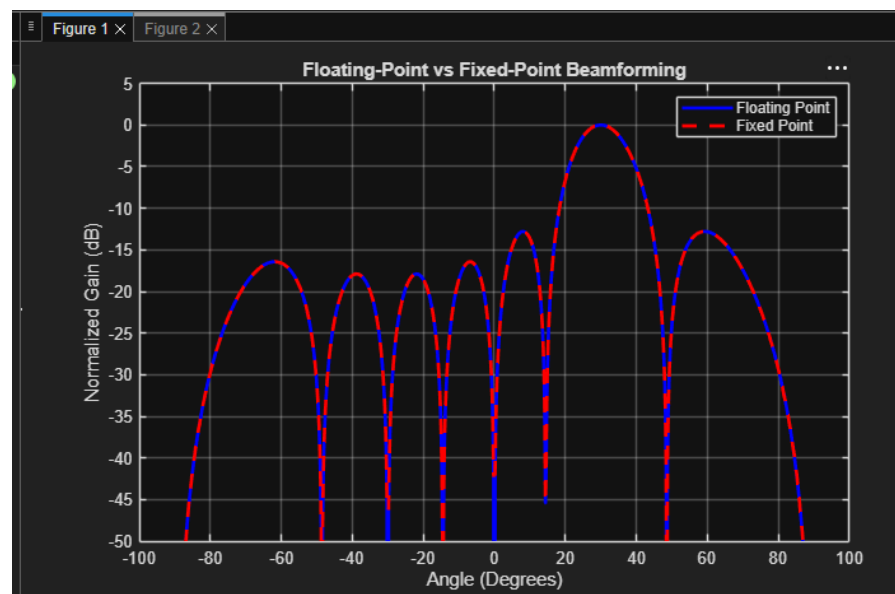
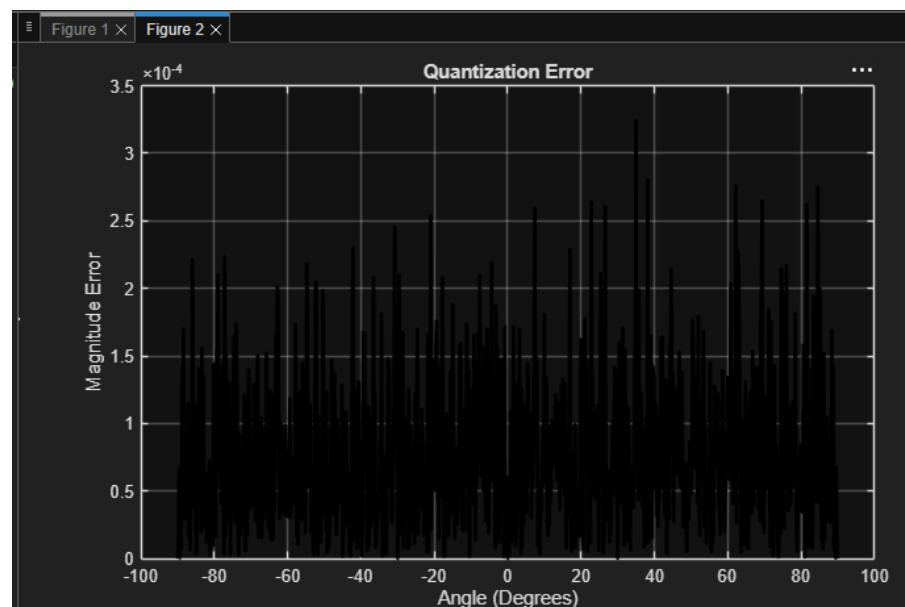


Figure 2: Quantization Error



Result

The floating-point and fixed-point beam patterns are nearly identical.

Quantization errors are very small:

- $\text{MAE} = 0.00007631$
- $\text{Maximum Error} = 0.00032438$
- $\text{RMSE} = 0.00009537$

Hence, the fixed-point implementation is recommended for FPGA/ASIC and real-time communication systems due to its lower computational complexity and memory usage while maintaining comparable beamforming performance.